# Yadlapally Shreedhanvi

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 8.00 |
| FIITJEE Jr. College | Class XII (TSBIE) | 2023 | 97.80% |
| Niraj Public School | Class X (ICSE) | 2021 | 96.33% |

## SKILLS

**Languages:** Python, C/C++, SQL, Java, TypeScript

**Tools:** Git, Github, Jupyter Notebook, Streamlit, LaTeX, Linux

**Frameworks & Libraries:** Numpy, Pandas, Matplotlib, Pytorch, CVXPY, FastAPI

**Competitive Programming:** Solved 500+ problems across Codeforces and LeetCode

## RELEVANT COURSES

**Artificial Intelligence:** Introduction to modern AI, Programming for AI, Foundations of ML, AI for Humanity, Deep Learning, Convex Optimization

**Computer Science:** Introduction to Programming, Data Structures, Algorithms, DBMS-I, OS-I

**Mathematics:** Discrete Mathematics, Probability and Random Variables, Matrix Theory, Introduction to Statistics, Numerical Analysis, Calculus

## PROJECTS

**Vision Foundation Models for Industrial Defect Detection** (Personal Project):

*Python - PyTorch - DINOv2 - LoRA*

- Developed a modular benchmarking framework for industrial defect detection, comparing CNNs and Vision Foundation Models through manual LoRA adaptation, achieving **0.92 AUROC** while training only **0.66%** of model parameters and evaluating deployment metrics including latency and memory footprint.
- Designed reproducible few-shot training pipelines with configurable experiments, feature-space visualization, and systematic benchmarking on the MVTec AD dataset.

**Sequence Learning from Scratch** (Course Project):

*Python - NumPy - PyTorch - Deep Learning*

- Implemented Vanilla RNNs, LSTMs, and Backpropagation Through Time (BPTT) from scratch in NumPy for sentiment classification, incorporating trainable embeddings and gradient clipping.
- Built and evaluated CNN and Transformer-based models in PyTorch, comparing architectures, optimizers, regularization techniques, and sequence lengths across image and text classification tasks.
- Analyzed model performance through systematic experiments and ablation studies.

**MiniML Machine Learning Library** (Course Project):

*Python - NumPy - Machine Learning*

- Developed a modular machine learning library with a scikit-learn-style API, implementing **10+** reusable components including L2-regularized Logistic Regression (IRLS), preprocessing pipelines, cross-validation, feature engineering, and model selection.

- Built an automatic differentiation engine and neural network framework from scratch supporting computational graphs, reverse-mode backpropagation, SGD optimization, and modular layers (Linear, ReLU, Sigmoid, Sequential).
- Benchmarked 5 model families across SpamBase and Fashion-MNIST, performing hyperparameter tuning with cross-validation and evaluating classical ML and neural networks using reproducible experimentation pipelines.

**Binary Image Classification for Kaggle Competition** (Team Project):

*Python - PyTorch - CNN - Computer Vision*

- Collaborated in a three-member team to develop a CNN-based image classification pipeline, securing **21st place out of 60 teams** on a Kaggle-style leaderboard through iterative model refinement and ablation studies.
- Designed the baseline end-to-end training pipeline, including data preprocessing, model training, validation, hyperparameter tuning, and reproducible experimentation.
- Identified model calibration issues and improved decision quality through threshold optimization, contributing to the team's final model selection and evaluation.

**Concurrent Producer-Consumer System** (Course Project):

*C - Concurrency - Synchronization*

- Built concurrent multithreaded and multi-process Producer-Consumer implementations using POSIX threads, shared memory and atomic semaphore synchronization.
- Implemented a synchronized circular buffer to coordinate multiple producers and consumers while preventing race conditions and ensuring data consistency.

## POSITIONS OF RESPONSIBILITY

**Sponsorship Head, Elan & nVision** (Annual Techno-Cultural Fest of IITH):

- Leading a team of 15 coordinators to acquire corporate sponsorships across diverse sectors, preparing proposals and negotiating deliverables with cross-functional teams and corporate stakeholders.

**Head Internship Coordinator, Office of Career Services:**

- Coordinating departmental internship activities, including recruiter outreach, internship drive strategy, communication, and execution logistics.

## EXTRACURRICULAR

Head of college's Indian Classical Dance Society.

Trained in Kuchipudi, had 100+ stage performances. NSS Volunteer.